Unit: Unit 0: Foundations of Algebra
Topic: Isolating Variables

Name: ____________________
Date: ____________

Foundation (Jalapeno)

Q1. Solve for x: $2x + 5 = 11$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q2. Isolate y in the equation $y - 3 = 2x + 1$
(A) $y = 2x - 2$
(B) $y = 2x + 4$
(C) $y = 2x - 4$
(D) $y = x + 2$
(E) $y = x - 2$

Q3. Rearrange the formula $d = rt$ to solve for t
(A) $t = \frac{d}{r}$
(B) $t = dr$
(C) $t = \frac{r}{d}$
(D) $t = r - d$
(E) $t = d + r$

Q4. Solve for v in the equation $v + 2 = 3t - 1$
(A) $v = 3t - 3$
(B) $v = 3t + 1$
(C) $v = t - 1$
(D) $v = t + 1$
(E) $v = 2t - 1$

Q5. Isolate x in $6x - 2 = 4x + 10$
(A) $x = 4$
(B) $x = 6$
(C) $x = 3$
(D) $x = 2$
(E) $x = 1$

Q6. Solve for F in $F = \frac{9}{5}C + 32$ for C
(A) $C = \frac{5}{9}F - \frac{160}{9}$
(B) $C = \frac{5}{9}F + \frac{160}{9}$
(C) $C = \frac{5}{9}F - 32$
(D) $C = \frac{5}{9}F + 32$
(E) $C = 9F - 160$

Q7. In the equation $a = \frac{G \cdot m_1 \cdot m_2}{r^2}$, solve for G
(A) $G = \frac{a \cdot r^2}{m_1 + m_2}$
(B) $G = \frac{a \cdot r^2}{m_1 - m_2}$
(C) $G = \frac{a \cdot r^2}{m_1 \cdot m_2}$
(D) $G = \frac{a \cdot m_1 \cdot m_2}{r^2}$
(E) $G = a \cdot r^2$

Q8. Solve for r in $A = \pi r^2$
(A) $r = \sqrt{\frac{A}{\pi}}$
(B) $r = \sqrt{A \cdot \pi}$
(C) $r = A \cdot \pi$
(D) $r = \frac{A}{\pi}$
(E) $r = \pi \cdot A$

Open Ended Questions (FRQ)

Q9. The equation for the force of gravity between two objects is $F = G \cdot \frac{m_1 \cdot m_2}{r^2}$. Solve for m_2 and show your work.

Q10. The speed v of a celestial body can be found using the equation $v = \sqrt{\frac{G \cdot M}{r}}$, where G is the gravitational constant, M is the mass of the body, and r is the distance from the center of the body. Solve for M.

Chef's Tips

- Ensure students understand the concept of isolating variables
- Remind students to check their work and units
- Provide examples of literal equations from physics for application